1. **Title of the Project: Development of an Integrated Intelligent Disaster Management System for Mountain Communities in India**

Principal Investigators: *(*Name, Designation and Affiliation)

	PI 1	**PI 2**	**PI 3**	**PI 4**	**PI 5**
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Age:	38	43	**43**	45	45
Category (GEN/OBC/ SC/ST)	GEN	GEN	GEN	GEN	OBC
Gender	MALE	Male	Male	Male	Female

3. Nature of the Project (*please mention the area)***:**	
1. Technology Development in association with an industry/ user agency.	√

2. Laboratory scale demonstration of a process/ device	√
3. Pilot plant demonstration for techno-economic analysis	
4. Nationally co-ordinated project to develop a specific technology platform	√
5. Development of a base technology and creation of a national facility	
6. Development of a technology of societal importance with commercialization plan.	
7. Field level demonstration of a developed technology for social acceptance	
8. Development of a cutting-edge technology relevant to the to specific industrial sector	

4. Duration of the project: - (Upto 36 months only) 36 Months

5. Names of participating Organizations with their addresses

Government Agency
National Institute of Disaster Management, Govt. of India, New Delhi 110042

Academic Institute
Indian Institute of Technology Indore, Indore MP, 453552

Non-Government Organisation
VARISHT(Voluntary Association of Researchers in Indigenous Knowledge, Science and Spirituality, Technology and Homeland Security), M382 A(firstfloor) Bharati villa, Ashiana colony, Lucknow-226012 Email:varishtglobal22@gmail.com.

Industry Partner:
Tata Consultancy Services, (TCS), TRDDC, Hadapsar I.E. / Sahyadri Park (SP2), Hinjewadi, Pune, India.

6. Objectives of the Proposal *(precise and preferably quantified process parameters/ product specification etc. in the bullet form)**:***

The goal of this project is to develop an intelligent responsive system based on a comprehensive knowledge database built on disasters, traditional knowledge of mountain communities, real time monitoring sensors, spatial information systems, and auxiliary sources of data focusing on multi-hazard early warning and management in hilly regions of

India. More specifically the objective is to support AI based tools for knowledge extraction and decision making for efficient coordination in disaster management and response. Thus, following are the broad objectives of the project

- Implement frameworks for obtaining static and dynamic data from different source types like hazard monitoring sensors, laboratory model test results (rainfall threshold and critical acceleration), traditional knowledge from local communities, spatial information sources, and web sources providing weather, news, and social media reports.
- Design and development of Intelligent Wireless Sensor Probe (IWP) for capturing multi-hazard parameters.
- Develop automated techniques for processing, refining and fusing multimodal data obtained from these sources to create a hazard data lake.
- Develop AI and ML based models for hazard mapping and susceptibility indexing, knowledge extraction for event prediction, early warning thresholds, real time situation awareness, event identification, and timeline generation.
- Develop a unified evaluation framework for AI based models targeted towards the above mentioned goals.
- Replicate real field scenarios in the laboratory based on the available geological/geotechnical data and topography of a vulnerable site and simulating rainfall and earthquake conditions to validate early warning of landslides due to combined effect of rainfall and earthquakes
- Develop appropriate use cases for leveraging project outcomes, identification of evaluation markers, and establishment of technology at laboratory scale and deployment of proof of concept. Focus area will be selected sites of Uttarakhand.

7. Fulfillment of Technology Qualifiers' Criteria:

Please mention if the proposal is

based on established R&D outcome/ results	
conforming to national/ international specifications	
potentially useful, demand driven and required by other agencies and users	√
a development of technology for multiple applications	
an adaptation of existing technology for its applications other than originally intended	
meeting a critical national need (present/ future) and strengthening technological capabilities for the same	√
an application of advanced science and technology with a promise of giving competitive solutions	

8. Cost Benefit (C-B) Analysis)/ Cost-Economic Analysis Report: -

Disaster management requires a multi-pronged holistic approach for monitoring and early

detection along with dedicated manpower and other resources for both decision making and on the ground activities for preparedness, response, as well as post disaster recovery. Monitoring and early detection would help in better preparedness thereby reducing loss of lives and infrastructure damages that directly affects the economy. Thus upgrading existing monitoring techniques with state of the art mechanisms is a prime necessity in any disaster management task. Furthermore, while decision making is based on both past experience as well as current ground situation reports, obtaining both long term experience as well as the current situation during any disaster may be challenging. According to the figures published by Statista research department in 2022, India has spent more than 300 billion Indian Rupees for relief in consecutive years 2016 and 2017. As disaster situations demand quick response and rehabilitation efforts by the administrator, quick and proper decisions in terms of resource distribution and workforce mobility is required. Improper decisions based on insufficient or incorrect information may incur huge wastage and loss to the exchequer. According to an annual report issued in 2012 in US by their Justice Department's Disaster Fraud Task Force it was mentioned that in cases related to Hurricanes Katrina, Rita and Wilma alone, the task force through FY 2011 prosecuted 1,439 individuals in 47 federal districts throughout the country for improper usage of disaster relief funds.

The situation is no different in India where several media reports were published regarding inaccuracies in fund distribution after several natural disasters. The major goal of the data-driven techniques for disaster management is to facilitate the administration of decision making through automated means by gaining quick knowledge from the past disasters as well as obtaining a more precise view of the current situation post disaster. This would help in better decision making and would mitigate the risks involved in distribution of relief funds, in terms of wastage, scams and fraud. The project aims to build an AI enabled integrated and scalable system with minimal cost involved that delivers the best performance. The system will provide the researchers to automatically evaluate their models using the benchmark data and thus the best performing model would be tracked. Thus the system auto-upgrades itself and provides the best support to administrators in decision making.

9. **Role of Industry (If any):-** Industries that work on disaster related systems may play a very important role in the project by improving its techno-commercial aspects. The role of the industry would be to bring in its expertise in developing commercial products ensuring that the developed product is more market driven and meets the industry standard in terms of the quality. This would enhance the commercialization prospect and reduce the time to market. Further, partnering with the industry to leverage their expertise in commercial system development is expected to mitigate several unforeseen risks in the project. Finally association with the industry would help in providing better long term post delivery support and maintenance, which is necessary to utilize the outcome fully. Hence, the IIT Patna team has partnered with Tata Consultancy Services (TCS).

   TCS has been actively involved in industrial disaster management related research and would be playing an important role in system requirement specification as well as in the design and development phase. TCS has done relevant research work in developing machine learning and NLP based models for information and knowledge extraction,

identifying events, entities, relations among them as well as frequent patterns from historical industrial safety incident reports. TCS has also worked in knowledge formalization and its computationally amenable representation to facilitate further reasoning and spatio-temporal analyses such as timeline generation, identification of similarity incidents and identification of recommended actions/suggestions given in previous incident reports, etc. These models have been used to analyze and suggest preventive measures for the industrial incidents and disasters. TCS would be playing an important role not only in detailing the system requirements and specifications, and in the design and development phase, but would also help in delivery and commercialization of the product. Furthermore, more importantly, TCS is providing support in terms of manpower (through paid internships) to be leveraged for the project.

NITK has partnered with SGES Pvt. Ltd a Geotechnical Engineering company with global presence. SGES is committed to contribute to the project both in kind and in cash with Rs. 15,00,000. They have also committed to contribute towards the physical model testing and for the technology transfer.

10. Origin and Justification of the Proposal: -

(Please justify based on the identified Qualifiers' criteria)

The increasing trend of extreme hazard events and climate-induced disasters over the past few decades is evident that climate change is happening. The consequences of climate change are directly or indirectly leading to the many vulnerable situations, thus causing disasters. For ages, communities in Indian villages have followed certain traditional wisdom sutras based on indigenous knowledge systems to predict certain weather and climate induced disasters that may cause significant damage to crops, livestock, and other aspects of village life. Village folk routinely followed such traditional, observation-based predictive models to avoid the high economic costs and disruptions to their livelihood and overall stability to the order of life. However, due to migrations, break in the continuity of the traditional systems, etc. with its cost-effective and widespread usage, such knowledge has been going out of vogue. Other modern systems have not effectively substituted this to the desirable extent. Local knowledge is already considered part of the Hyogo Framework for Action (HFA) Priority 3 which emphasizes using traditional and indigenous knowledge to help mainstream disaster mitigation policies and practices. Sendai Framework for Disaster Risk Reduction (2015-2030) focuses on developing a more people-centered prevention approach to prevent new risk and minimize emerging risk, thereby safeguarding people's lives, livelihood, and assets. Despite having much emphasis on integrating indigenous knowledge in Disaster Risk Reduction(DRR) and Climate Change Adaptation(CCA) policies and practices, the increasing number of projects and initiatives being undertaken that consider natural environment as a crucial issue and placed it in strategic frameworks. Many humanitarian factors, the scientific community, policymakers, built environment professions still feel there is a lack of shared understanding about issues such as who is responsible for integrating indigenous knowledge to inform the strategic directions in disaster response. How

to practically do it, which tools, services are available, and in what ways these tools can be cross referenced. There have been many scientific advancements in Artificial Intelligence-driven tools, methods, and systems developed to mitigate disaster risks and generate heat maps of the risk in recent decades. Nevertheless, many such tools are limited to secondary and spatial data. There have not been sufficient models or proven methods that have tried to integrate the local and indigenous knowledge from the communities using Artificial Intelligence tools and techniques. The reason for this is that Indigenous knowledge and risk are place specific. This project intends to develop an advanced, cost-effective, technology and AI driven solution that leverages and integrates the traditional and tested indigenous knowledge of the hill communities that can help predict and mitigate the short and long-run impacts from climate-induced disaster risks.

11. Summary Outline of the Project (with Schematics, where possible; Define the Problem and give technical details)

This project aims to create a multi-hazard early warning, decision making and management system that is intelligent and responsive and is based on a comprehensive knowledge database on disasters, traditional knowledge of mountain communities, real-time monitoring sensors, spatial information systems, and auxiliary sources of data. The outcome of the project would be leveraged not only by government agencies for decision making and providing citizen-centric services, it would also be leveraged by both developers and researchers for scientific study and to build robust tools and platforms for disaster handling and management. The figure 1 illustrates the high level functionality of the system.

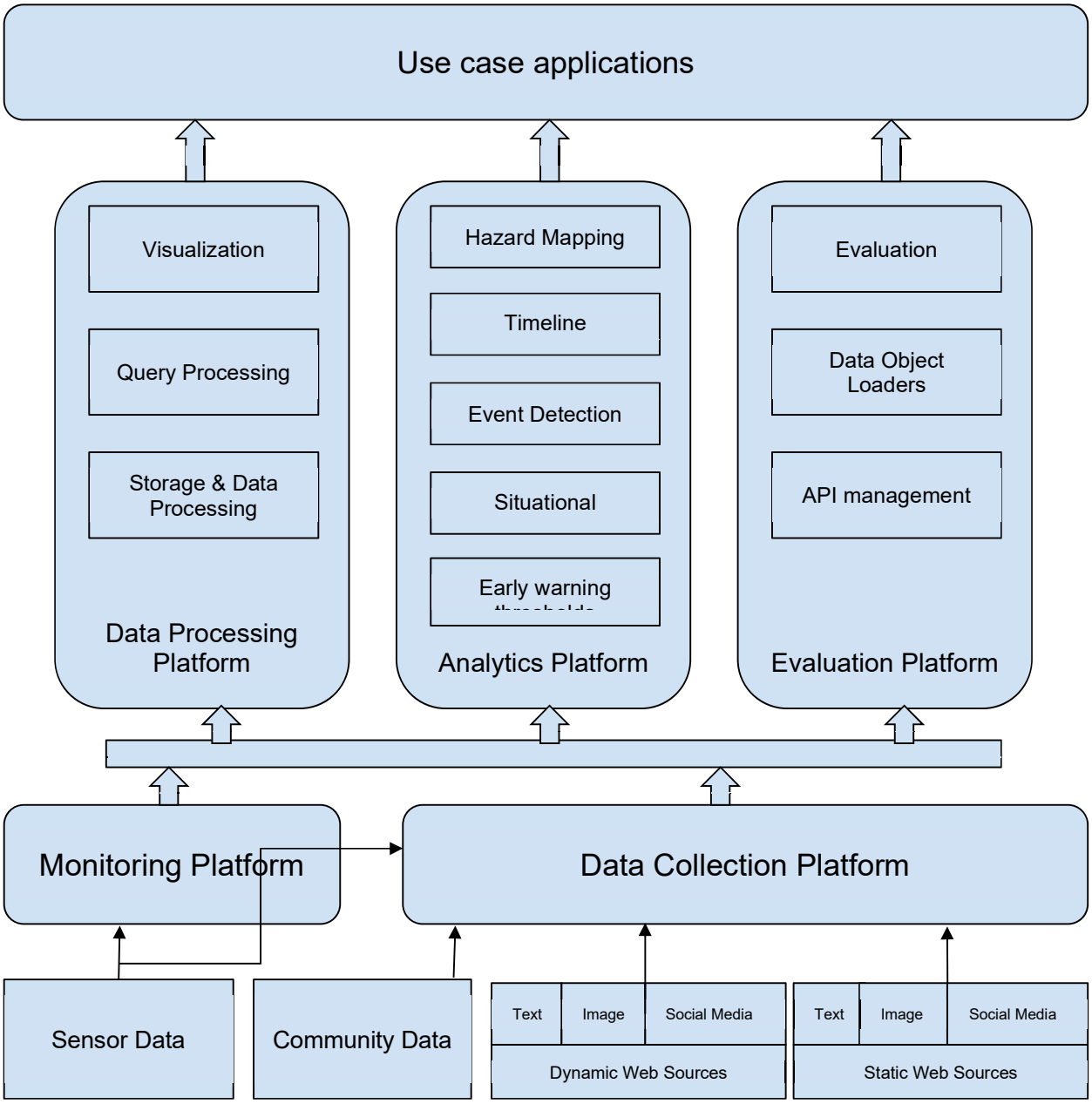


Figure-1: Functional components of the system

We aim to target this objective by developing a setup with six major functional requirements:

1. Creating a framework to collect multimodal data from different source types like hazard monitoring sensors, laboratory model test results (rainfall threshold and critical acceleration), traditional knowledge from local communities, spatial information sources, and web sources providing weather, news, and social media reports. The data from web sources may be dynamic or static. The sources for dynamic data include images and videos from news and meteorological sites, news articles published in different languages, and streaming multimodal data from social

media feeds. On the other hand, static data sources include past incident and damage reports, images and videos in news and weather repositories, published news articles, and collected social media feeds. The data from both dynamic as well as static sources would be obtained through standard APIs and libraries that are publicly available. This data forms the backbone of the data-driven disaster management services and would be leveraged for various services to be provided by the system.

2. Create an intelligent and integrated wireless sensor probe for real-time monitoring and communication for early prediction and decision making.
3. Create a platform for processing, refining and fusing multimodal data obtained from these sources to create a hazard data lake that can be used for different AI related tasks. The collected raw data would initially be refined using standard methods of deduplication, denoising, and handling of missing values. Subsequently, novel AI based techniques for multimodal data fusion techniques would be used for extracting information from different data modalities and creating a common data representation. This would help in creating more complete benchmark datasets that can be leveraged for training different AI based models. Benchmarking the datasets would involve both automated as well as human annotation techniques. Furthermore, this platform would also be used to replicate real field scenarios in the laboratory based on the available geological/geotechnical data and topography of a vulnerable site and simulating rainfall and earthquake conditions to validate early warning of landslides due to combined effect of rainfall and earthquakes.
4. Leverage the benchmark data for developing AI based models for key services like hazard mapping, knowledge formalization, early warning thresholds, real time situation awareness, and event timeline generation of past disasters that can be used for decision making. The timeline generation process would involve extracting the major events from static data sources, followed by temporal ordering of the events that would provide a holistic overview of the important micro-incidents in past disasters and the actions planned to handle them.
5. A major requirement is to develop a unified model evaluation platform on top of the framework that can be used to compare newly developed AI based models with existing models submitted to the system. This would involve creating different APIs and data loaders along with identification of evaluation markers for comparison and evaluation.
6. Finally develop appropriate use cases for leveraging project outcomes, identification of evaluation markers, establishment of technology at laboratory scale and deployment of proof of concept. Focus area will be selected sites of Uttarakhand.

12. Expected Outcome in Physical Terms:-

a. New/ Upgraded Product - Yes
b. New/ Upscaled Process- Yes

1. An Artificial Intelligence (AI) based integrated technology/hardware system comprehensive response and recovery system for Disaster Management. The system would involve

seamless integration of Spatial Information, High Resolution Cameras, Unmanned Aerial Vehicle (UAV), High Speed Data Processor, Relays and Communication Systems for multidirectional flow of data for faster and effective responses to disasters by various concerned agencies.

2. A new integrated product to be used in large scale natural and manmade disaster scenarios for
    a. Automated data collection from static and dynamic data sources
    b. Real time situational awareness
    c. Event identification
    d. Timeline generation of past disaster events
3. Technical reports on novel processes proposed for
    a. Multimodal data fusion
    b. Creating multimodal benchmark data from heterogeneous sources
    c. Task specific information filtering for situational awareness
    d. Information retrieval for event identification and timeline generation
4. An unified evaluation system for testing AI based models related to
    a. Task specific information filtering
    b. Event detection
    c. Timeline generation
5. A set of labeled multimodal benchmark datasets for training models related to
    a. Task specific information filtering
    b. Event detection
    c. Timeline generation
6. Trained manpower with knowledge in
    a. AI and Machine Learning
    b. Big data technologies
7. Develop a compatible framework to document and archive the diverse traditional knowledge systems of hill communities that can eventually feed into the AI-engine.
8. Database of Cultural maps indicating traditional knowledge systems, risk maps its knowledge resource network.
9. Documents different adaptive practices of local communities and identify their common and specific trends in terms of their changing behaviours and living practices.
10. Deterministic and probabilistic earthquake hazard maps with latest data and models
11. Landslide vulnerability maps considering 12 geological parameters, rainfall data and earthquake acceleration
12. Lab-based physical model for early warning of landslides with combined effect of rainfall and earthquakes
13. Selection of most vulnerable location and deployment of rain gauges and sensors in selected vulnerable locations
14. Establishment of real-time communication from the sensors to the central location
15. Development of AI and ML based threshold models for early warning of disasters.

13. Targeted Specifications of the Expected Outcome: -

1. Timely and efficient response in disaster prone areas. Efficient resources management.

2. The targeted specifications are different for different outcomes and are mentioned as follows:

I. Data collection framework
   a. Supported static data sources should include historical incident and damage reports, images and videos from news and weather repositories, text from multilingual news articles, and past available social media feeds.
   b. Supported dynamic data sources should include downloadable images and videos from various news and meteorological sites, texts from news articles, published incident reports, and social media feeds like Twitter, Reddit and Facebook.
   c. Supported languages should be English
   d. Should support all standard image and video formats, html and xml formats, and is on the format for social media feeds

II. Analytics framework
   a. Situational awareness in terms of
      1. Affected location names
      2. Damage score
      3. User requests and needs
      4. Injuries and deaths
      5. Response to evacuation
   b. Event identification and Timeline generation
      1. All major natural and manmade disasters in the last 5 years
      2. Graphical visualization

III. Data repositories
   a. Separate repositories for image and video data, web articles and social media feeds
   b. Suitable anonymization of all data

IV. Unified evaluation platform
   a. Provides data objects for labeled data compatible with python libraries
   b. Should support codes written in python for task specific information filtering, event identification and timeline generation

3. The targeted specifications of the expected outcome supported by intelligent wireless probel (IWP):

I. Assessment of vulnerable location in terms of the population, infrastructure and other elements that are directly under risk.
II. Deployment of rain gauges maximum 2 numbers at specific locations
III. Deployment of other sensors such as moisture sensor, pore water pressure sensor, movement sensor, Geophones etc to capture real-time landslides triggering vital parameters.
IV. Establishment of communication standards like wifi, Zigbee, cellular networks

depending upon the best options available at the deployment location.

V. Developing AI and ML based thresholds from the historic and real-time data to be used for early warning of diasters.

4. The various points under geological and geotechnical considerations:

I. Detailed assessment of earthquake hazard using deterministic and probabilistic approaches

II. Geological and geotechnical considerations for preparing Landslide vulnerability maps induced from both rainfall and earthquakes considering Frequency ratio method and Newmark analysis

III. Lab-based physical model for early warning of landslides with combined effect of rainfall and earthquakes

14. Deliverables of the Project *(precise and in bullet form):-*

The following are going to be the deliverables of the project:

1. Deployment of sensors with real-time communication to understand the real-time risk from selected vulnerable locations.
2. A laboratory scale AI based comprehensive response and recovery system for disaster management that will be tested through simulation studies and available dataset.
3. An AI based integrated technology/hardware system would involve seamless integration of High Resolution Cameras, Unmanned Aerial Vehicle (UAV), High Speed Data Processor, Relays and Communication Systems for multidirectional flow of data for faster and effective responses to disasters by various concerned agencies.
4. A data collection framework for collection of disaster related data from
    a. Digitized incident reports
    b. Multilingual websites (we target English, Bengali and Hindi)
    c. Images and videos from news and weather repositories
    d. Social media feeds like twitter and reddit
5. Technical reports on novel mechanisms for
    a. Multi-modal data fusion
    b. Creating multimodal benchmark data from heterogeneous sources
    c. Task specific information filtering from multilingual social media feeds
    d. Event identification and timeline generation from multimodal and multilingual web and social media data.
6. Benchmark datasets of
    a. Multimodal social media feeds classified as informative/non-informative, with

informative feeds further classified based on different humanitarian tasks like infrastructure damage, loss of lives, need, help etc., along with proper language tags.

   b. Images and videos classified as informative/non-informative

7. An unified evaluation system for testing AI based models related to
   a. Task specific information filtering
   b. Event detection
   c. Timeline generation
8. Develops the Cultural maps related to Risk perception, and Adaptive practices of the local hill communities.
9. Prepare the database of cultural resource network including tangible and intangible elements.
10. Evaluation of a location's vulnerability in terms of its population, infrastructure, and other directly vulnerable factors and selection of the most vulnerable location to deploy sensors.
11. Deployment of rain gauges in a maximum of two places deployment of additional sensors such as moisture sensors, pore water pressure sensors, movement sensors, geophones, etc. to capture and monitor critical and essential real-time landslide triggering variables
12. An established real-time communication standard after choosing from the best communication alternatives available at the deployment location.
13. Developing AI and ML based thresholds from the historic and real-time data to be used for early warning of diasters.
14. Deterministic and probabilistic earthquake hazard maps with latest data and models
15. Landslide vulnerability maps considering 12 geological parameters, rainfall data and earthquake acceleration
16. Lab-based physical model for early warning of landslides with combined effect of rainfall and earthquakes
17. Trained PhD students and publications in high quality journals and conferences

15. Methodology

(Please highlight how success in the project execution will be ensured)

1. Development of data collection and Ingestion Framework

This study adopts Geo Spatial Analysis, Real Time sensor-based monitoring, qualitative and quantitative methods from interdisciplinary subjects to develop a novel framework. For the

first time of its kind in the sensitive studies related to national disaster management in India. This study attempts to analyze rich and often distorted qualitative text (in the form of stories, observations, experiences) from traditional knowledge of local communities, sites with the understanding of social and adaptive behavioral patterns towards their understanding of their dynamic settings and their threat source (local/regional/external). Such analysis will define the possible threats to communities, society, environment, region and elsewhere for similar contexts. Along with threat classification, source of such threats and their patterns and paths would then be identified and mapped using GIS techniques. This would then show a way to understand the semantics associated with such qualitative information with the quantitative data sets and conclude on the possible future threats to societies, their habitat settings and their environments. Also, this would also help to identify the network of such Geotagged threats and their concerned vulnerabilities.

There are five components of the study that formulates as a methodological framework.

1. *Types of Data Sets involved :* There are three components of Data sets involved in the study. Field level quantitative and qualitative data, spatial data and desk based raw data collection. In field level there are two types. Real time and field level community based social.

2. *Real Time Data:* One of the field level data is the real time data monitoring. In real time data, multi-hazard detection sensors will be installed at selected villages and variances will be monitored and then compared with the perceptions of community. The design optimization, fabrication, testing and pilot field deployment of the proposed device for multi-hazard early warning of landslides and floods will be conducted in three phases:

    1. Design and Fabrication of the sub-modules including:
        a. integrated multi-hazard sensor system,
        b. Context-aware multi-source power system,
        c. Signal Conditioning and Preprocessing Board,
        d. Real-time Heterogeneous Fault Tolerant Communication system,
        e. Local-level warning generation system using edge analytics.
    2. Sub-system's bench level testing and calibration,
    3. Full System assembly and testing at multi-hazard Laboratory: Design Iteration and Optimisation with assistance of Industry Partners,
    4. Fabrication and Field Deployment of final device/product for real-world assessment.

3. *Ethnographic and Socio-spatial data from the local communities:* The second level of social data from the field is to document Indigenous/Traditional Ecological Knowledge that communities possess through their long experiences, observations. What form/time does this Traditional Ecological knowledge exists? Whose knowledge it refers to? How to document such diverse forms of knowledge objectively? How are such knowledge systems transmitted from one generation to another? What kind of threats do exist for such knowledge base? To address these questions following methods will be employed.

- Direct and indirect observation
- Oral Histories
- Semi-structured interviews
- Archival studies

Mind map analysis. Data collected exist in different forms, eg. transcribed texts, audio and visual recordings, and cognitive maps collected from the community. All such data will be analyzed through descriptive and interpretive frameworks. Following the transcription and identification of keywords, the data was sorted into domains and conceptualized to highlight the issues, their causes, and the impacts that have occurred in the spatial and non-spatial domains through stakeholders' lens.

4. *Geo Spatial data* is used to analyse the LULC patterns, drain patterns, Multi-hazard risk mapping, target group asset mapping, infrastructure related to disaster response will be mapped by using Geo spatial tools and Satellite imagery.

5. *Secondary or Raw data collection:* We divide data sources into 2 different categories, dynamic and static. For each category, we consider 4 major sources of data, images from news and weather sites (image data), videos from stored repositories, news reports in websites (static multilingual text data), and social media feeds (streaming multimodal data). The framework to collect each of these data for each of the categories needs to be developed and integrated. We limit the scope of the data to natural hazards itself.

Data mining is finding new information in a hidden or latent dataset. Data mining can help people better understand a large set of data. Supervised and unsupervised algorithms are used to identify the hidden patterns in data. Supervised approaches depend on some a-priori knowledge of the data (e.g. class labels). Unsupervised algorithms are used to characterize data without any prior instruction as to what kinds of patterns will be discovered by the algorithm (Barbier, 2011).

A Machine Learning pipeline will be built that will continue to churn out results as soon as data is generated and collected. Since we are dealing with Big Data, the programming language of our choice will be python (pySpark) which is powerful and can handle a continuous stream of data. The core Spark API will enable scalable and fault-tolerant stream processing of live data streams.

The streams of data from sensors have to be converted into discretized data by doing processing on the original data. The processing stage involves defining the batch duration for the data resource. According to the batch duration, the data will be stored in Resilient Distributed Dataset (RDD). The chain of continuous series of RDDs can be used as a distributed dataset by Spark. Hence, an organized dataset will be obtained.

2. Data Analysis through Artificial intelligence tools and methods

Dataset creation — refinement and fusion: For the data collected from static sources, the collected raw data would initially be cleaned followed by a multimodal data fusion. The cleaning step involves removal of unwanted or duplicate observations, fixing of structural errors, detecting and removing outliers and handling of missing data. The multimodal data fusion step would involve use of deep learning models for extracting the latent features from the multimodal data that would subsequently be leveraged for fusing related data. The following step is data cleaning which involves null value imputation. Then comes Exploratory Data Analysis. The valuable insights drawn from the data will then be used in tuning the parameters of the machine learning and deep learning models that will be built in the next stage. Data will also be visualized on the dashboard of the portal in real-time. Sensitive areas will be plotted on a heat map of the Region of Interest. An example of heat map generated through initial work (developed by Surya Remanan) is shown in figure 2.

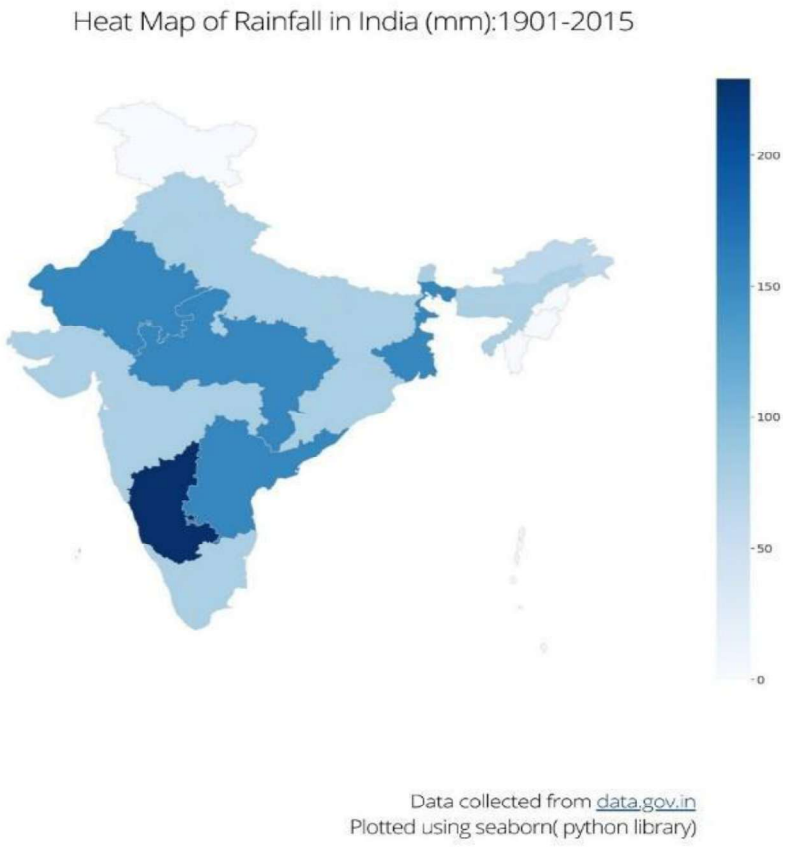


Figure 2: Sample Heat map of Rainfall in India

Information retrieval and filtering: The major goal of this work package would be to develop deep learning based techniques for information extraction, retrieval and filtering to

obtain situational information from data, obtained from dynamic sources. Models would be trained on static benchmark data for multimodal deep learning and subsequently applied on dynamic live data for real time situational awareness post disaster. The next stage is model building. According to the nature of data, be it numerical, textual or visual the algorithms will be designed. Supervised, Unsupervised, Semi-supervised or Self Supervised Machine learning algorithms will be designed to fit the use case and predict disasters.
In the case of numerical data collected from sensors, algorithms used will range from classical approaches to state-of-the art algorithms depending upon the situation. For different scenarios, we have to put into use different factors and parameters which will come into play.
In the case of qualitative data, data will be coded using content analysis or grounded theory method.

Cognitive computing Techniques: Simulating of human thought processes in a computerized model. Cognitive computing involves self-learning systems that use data mining, pattern recognition, and natural language processing to mimic human behavior. This is a valuable technique to predict future threats from the vivid variety of traditional and secondary data sets and to identify potential vulnerabilities for society, environment and Nation.
Event identification and timeline generation: The major goal of this work package is to develop techniques for identifying events and generating a timeline of the major events and activities for a past disaster from the corresponding static data. The objective is to extract knowledge and information from the past disaster experiences to facilitate decision making for the administration in various phases of a current disaster.
Developing appropriate use cases and the unified evaluation: There are 2 major goals of this work package. First is to develop appropriate use cases to leverage the project outcomes, and subsequently develop and deploy the proof of concept. Another goal is to develop the necessary tools for the unified evaluation platform, where users can use the data loaders provided through the platform to download the data generated and create data objects that would be compatible with standard NLP, image and graph libraries of python.

3. Developing a Geospatial interface:
This proposed project's state of the art lies in integrating traditional knowledge analysis with Geotagged spatial and secondary data for the systematic development and execution of data mining techniques, and developing a framework for further classification of threats based on the intended target sectors. After analyzing susceptible maps and patterns of hazards and identifying the potential vulnerabilities, Geotagged data along with future threat, alerts and source details will be entered into a database. The outcome of this project delivers a web GIS portal and app based by using free and open-source technology, ensuring the future development possibility from the most robust GIS open-source community. Developed user interface map threats as point feature and by using feature services alert the monitoring agency about the location of the threat. Such outcomes would help the decision-making authorities and support system network those including security agencies, government bodies, and business sectors on how to plan the required actions according to the threat and vulnerability analysis. In this way, all the three components of disaster management,

including mitigation, preparedness, and Response, get covered where correct planning will helps in mitigation. And while mitigating any situation, we will be learning certain things using geospatial technology that will prepare us for future planning to mitigate any disaster in the future.

4. Communication with the Stakeholders:

This project develops a self-learning AI based model that provides a two-way communication. It simultaneously communicates the alerts to communicate the local communities and other relevant stakeholders and at the same time, it will be collecting the collecting, learning, and processing the data. The outcome of this project is cost effective Application based Web GIS model that can be easily understood by the common man in mitigating the disaster impacts and supports the development of response strategies and operational aspects. Through this the whole projects conceives a through bottom-up process rather than a top-down mechanism. The data comes from the bottom most community level, and the communication of the outcomes and findings reaches to the end users and the local communities themselves.

5. Scaling up for Commercialization

This product thus developed is a community tested product and hence this can be repeated elsewhere in similar conditions and contexts. The learnings from this testing process would enhance the technical scope of the future research projects. Through this the team reaches out to the governmental and business industry to support the prototyping and commercialization process, and to serve as a cost effective and indigenous AI driven model.

The following flowchart in figure 3 explains three major research stages.

Stage 1 involves WP-1 Data mining techniques. This project draws data from three different sources, primarily through the documentation of existing Indigenous knowledge systems through oral traditions, mapping techniques, direct and indirect observation. Secondly, through the secondary through the resources. Third, the data will be collected through the real-time monitoring mechanisms.

Stage 2 involves WP-2 and WP-3 Data Collation, Data integration on one side and the later step involves, Data Evaluation and Analysis further directing it to the Data interpretation and prediction.

Stage 3 involves WP-4 and WP-5 the dissemination processes such as establishing communication mechanisms, alerts to the communities, Deployment of AI model through WebGis portal.

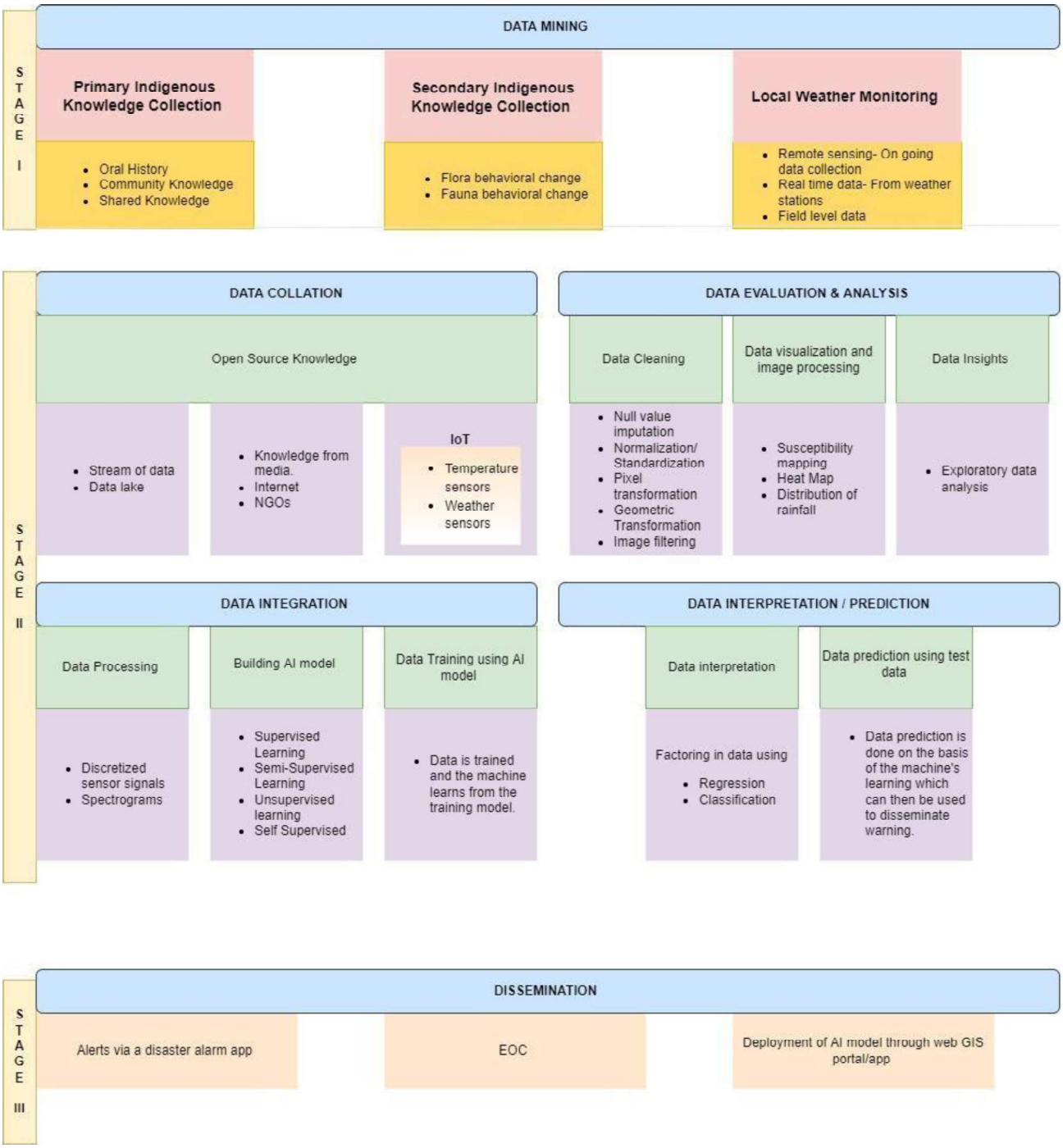


Figure 3 - Detailed Methodology chart explaining various stages

Following are the institute wise contribution to execute the project:

Activities and responsibilities of each institute

Work Package	Activity & Task	NITK	IITR	Amrita	IITP & TCS	UPES
WP1: Development of data collection and Ingestion Framework Lead:	IoT Edge Node Development			X		
	Sensor Data			X		X
	Traditional knowledge from local communities		X			
	Spatial Data	X		X		X
	Multimodal data from web sources				X	
WP2: Data refinement, pre-processing, labelling, data alignment and fusion Lead:	Sensor data			X		X
	Edge Computing			X		
	Traditional knowledge data		X			
	Spatial data	X		X		X
	Web source data				X	
	Automated techniques for multimodal data alignment and fusion			X	X	
WP3: Model Development Lead:	Hazard mapping	X		X		
	Preparedness indexing	X		X		X
	Knowledge extraction from community data		X		X	
	Situational awareness and damage assessment		X	X	X	X
	Major event identification from historical data			X	X	
	Disaster specific activity timeline generation			X	X	

	Real-Time Event Detection Models	X		X		X
	Real-Time Early Warning Models	X		X		X
WP4: Disaster Management & Services Lead:	WebGIS & Visualisation		X		X	
	Warning Dissemination					X
	API creation				X	X
	Report Generation				X	
WP5: Benchmarking and evaluation Lead:	Identifying critical features for assessing community specific vulnerabilities and risks		X			
	Identifying critical features for assessing post disaster response and recovery		X			X
	Identifying parameters for hazard prediction	X		X		
	Platform for automated evaluation of AI based models on damage and need assessment, event detection and timeline generation			X	X	
WP6: Use case development and proof of concept deployment Lead:	Proof of concept	X	X	X	X	X
	Use cases on hazard mapping and preparedness indexing	X		X		
	Use cases on community based studies		X			
	Use cases on damage and need assessment, and task planning				X	
	Use cases on assessing post disaster response and coordination					X
	IPR and Commercialization plans	X	X	X	X	X

16. Milestones with Dates, Work Elements for each and the Organization responsible for each Work Element

S. No.	Milestone	Target Date	Work Elements	Responsible Organisation
1	Cultural maps of Risk perception	Month 24	Fieldwork, Archives, and social media, mental mapping exercises, secondary literature studies	IIT R
2	Co-seismic and rainfall-induced landslide hazard mapping	Month 18	Earthquake hazard assessment Frequency ratio/Weighted average method	NITK
3	Landslide Flume Tests on shake table integrated with rainfall simulation using sensors	Month 30	Detect rainfall threshold, Detect pore pressure rise, Detect surface movement with LiDAR and camera	NITK
4	Manpower recruitment	3 Months	Review of available literature, Comparative assessment, Critical analysis	All
5	Critical analysis of best practices and identification of important parameters	6 Months	Brainstorming with stakeholders relevant for Uttarakhand, Traditional knowledge of prevention, Identification of Parameters and Protocols	UPES Dehradun

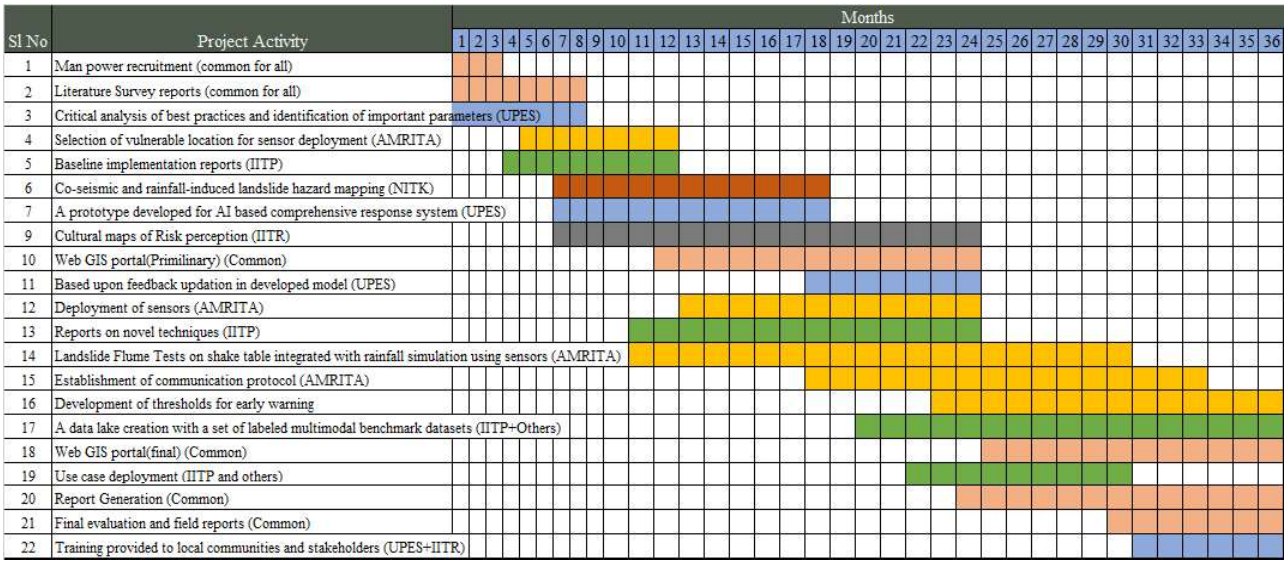
6	A prototype developed for AI based comprehensive response system	18 Months	Design of system with parameters and protocols, Simulation Studies, Testing on available dataset	UPES Dehradun
7	Based upon feedback updation in developed model	24 Months	Feedback, Correction and Finalization	UPES Dehradun
8	Reports generation	30 Months	Data collection, preprocessing, data fusion, identification and response techniques	All
9	Training provided to local authorities	36 Months		UPES Dehradun and IITR
10	Selection of vulnerable location for sensor deployment	12 Months	Traveling to fields, risk assessment at various locations and selection of a vulnerable location based on living population, infrastructure and elements under risk	Amrita
11.	Deployment of sensors	24 months	Purchasing sensors, testing in the laboratory and field test, selection of location for deployment, bore hole drills at	Amrita

			required locations and deployment	
12	Establishment of communication protocol	32 months	Testing of various communication standards in the field, Choosing the best alternate, establish the communication, streaming of real-time data from the field	Amrita
13	Development of thresholds for early warning	36 months	Developemnt of rainfall thresholds and other sensor based thresholds for early warning of disaters.	Amrita
14	Literature Survey reports	6 months	Data collection and refine, multimodal data fusion, information retrieval and filtering, event detection	IIT Patna
15	Baseline implementation reports	12 months	Multimodal data fusion, Information retrieval and Information filtering, technical report on proposed use case deployment: planning and evaluation markers	IIT Patna
16	Prototype development	18 months	Data collection framework, Visualization framework	IIT Patna, TCS

17	Reports on novel techniques	24 months	Multimodal data fusion, information retrieval and filtering, timeline generation	TCS, IIT Patna
18	Use case deployment	30 months	All modules	IIT Patna
19	Final evaluation and field reports	36 months	Integrated platform	IIT Patna, TCS
20	Web GIS portal	30 Months		All

17. Work Plan

(Provide a bar chart giving project activities and milestones. Highlight milestones)



Benchmarks to be achieved *(In quantitative terms)*:

1. Data collection framework
    a. Should be able to collect at least last 5 years of data from sensors, community level, web, satellite images and social media feeds
    b. Should store last 5 years of data in all formats
    c. Should be able to handle streaming data like real time sensors and tweets
2. Proposed processes
    a. All AI based models should be able to achieve sufficiently high accuracy and F1 scores.
    b. Information filtering mechanisms should have at least 2% better F1 scores as compared to the best performing state of the art baselines

c. Information retrieval mechanisms should have at least 2% better accuracy and weighted F1 scores as compared to the best performing state of the art baselines.

3. Dataset
    a. Benchmark dataset should be multimodal, consisting of sensor data, images, videos and text
    b. Labeled text dataset should have no more than 1% duplicates
    c. Labeled social media data should contain at least 10 events that occurred within last 5 years
    d. Manually annotated datasets should have at least 85% Kohen Cappa mutual agreement score

18. Novelty/uniqueness of the proposal (not more than one page):

The project's main innovation is the creation of a comprehensive data collection framework for collecting relevant multimodal hazard data from various sources, including sensors, local communities, GIS, and online sources. This data will then be used to create a data lake that will be used by the administration to coordinate various disaster management tasks and offer user-centered services. In order to improve its quality and scope, it also seeks to support academics and developers with benchmarking and assessment. The major innovative components include:

1. Creating a hazard data lake with heterogeneous data collected from diverse sources to facilitate analytics and AI based support tools.
2. Developing an Intelligent Wireless Sensor Probe (IWSP) which is Self-sustaining, Context-aware, and offers Adaptive Service Delivery
3. Providing holistic solutions for coordinating various disaster management tasks leveraging domain specific expertise from various disciplines, which otherwise works in silos.
4. Exploiting traditional knowledge gained from local communities for better planning and decision-making based on location and community-specific requirements and cultural practices.
5. ML and AI based decision models for detection and early warning of multi-hazards
6. Physical model tests to validate a real field scenario considering geological/geotechnical parameters and rainfall & earthquake vibrations
7. Developing a novel automated benchmarking and evaluation platform for AI based developers and researchers working on various disaster management tasks.

19. Gaps to be covered through proposed work with special reference to the proposal:

While traditionally automated disaster management systems have relied on sensors and satellite imagery for monitoring and damage assessment, newer sources of information like intelligent sensors, community knowledge, and social sensors provide rich source of information that may be exploited for better knowledge extraction. In this project we develop a comprehensive solution that encompasses knowledge from diverse areas of expertise including traditional knowledge systems

to build a robust and intelligent disaster management system. Further, although state-of-the-art AI models can provide better decision making capabilities, there is no standard benchmark data available for the same that can be used to train the models. This project aims to fill in this major gap. On the other hand, due to lack of benchmark data, existing models are evaluated on different crisis data collected in isolation. We provide a unified evaluation platform for the models so that the models are evaluated on a set of standard benchmark data and the best performing model is leveraged for the corresponding task.

20. Critical Review of latest Status of the Technology (maximum 2 pages each & with complete references):

21.1. National Status Review

National Disaster Management Authority (NDMA) conceived a national project, **"COMMON ALERTING PROTOCOL BASED INTEGRATED ALERT SYSTEM (SACHET)"**. The project intends to integrate the Alert generating agencies, alert dissemination agencies, and the State Disaster Management Authorities (SDMAs) on a CAP-based platform. A pilot project was carried out in Tamil Nadu successfully and pan India implementation of this project is under process. Another ongoing project by NDMA is about creating a single distress number in case of all emergencies across India[1].

Some studies have been carried out in the disaster management of India where machine learning methods are used in predicting disasters in the hilly and mountainous regions. One such example is about the snow avalanche sites in Leh Manali Highway. They used machine learning methods to predict snow avalanche sites in the region (Tiwari, et al., 2021). Another example where we find application of AI is during the cyclone Gaja (2018) in India. A GIS-based mobile application was developed. The app helped rescue workers, decision-makers, and other personnel carry out post-disaster emergency response efficiently (Sharma, et al., 2020). Further there are innovations by students where they developed an AI based 'Eye in the Sky' for disaster relief and aid purpose. Students developed this at IIT Madras, they use drones to collect data and analyse it in real time. This helps identify people trapped in disaster hit areas and inform them of the relief task force[2].

In India, many indigenous communities possess tremendous knowledge about their local environment and helps in predicting and managing disasters. For instance, the tribal communities in Rajasthan (Bhil, Meena, Kathodi to name a few) have knowledge on natural disasters and extreme weather events (Pareek & Trivedi, 2011). This knowledge base is used in many areas like biodiversity conservation, forest management and so on, but this knowledge and AI are seldom used. Incorporating this knowledge base and AI can improve efficiency in disaster

[1] https://ndma.gov.in/Capacity_Building/Ops_Comm/IT_Comm_Project

[2] https://www.livemint.com/news/india/iit-madras-students-develop-ai-based-eye-in-the-sky-for-disaster-relief-aid-11570622440608.html

management.

21.2. International *Status Review*

Artificial intelligence is being used in multiple fields and is used to solve challenges and improve systems. AI is also applied in predicting natural hazards in many regions and supports informed disaster management. There are four phases in disaster management: mitigation, preparedness, response, and recovery. Multiple AI methods are in use for different application areas under different phases.

Globally, AI is extensively used in disaster management. For instance, in the region of southwestern Iran, the probabilities of natural hazards like snow avalanches, landslides, wildfires, land subsidence, floods are predicted using machine learning models. Namely, support vector machine (SVM), boosted regression tree (BRT), and generalized linear model (GLM) (Yousefi, et al., 2020). Another example of AI used for landslide prediction and mapping is in Shicheng County, Jiangxi Province of China. A comparative study was carried out between different models to determine which model can be used accurately to predict and map the landslides in the region (Huanga, et al., 2020). There are other studies where multiple machine learning (ML) and Convolution Neural Network (CNN) methods were used to map and predict landslides in the Himalayan region in Nepal (Ghorbanzadeh, et al., 2019). There are multiple AI algorithms which are used to detect other natural hazards like earthquake prediction, food vulnerability mapping and plotting, flood and fire management, post-earthquake mapping to name a few (Abid, et al., 2021). Further, there are few literary sources where we can see that traditional knowledge of indigenous people and AI are used for disaster management. Traditional ecological knowledge is nothing, but knowledge accumulated over a certain period by a specific group of community about their environment and land management. This knowledge is passed down from one generation to the next by word of mouth. One such case is about the Canadian Aboriginal groups in their forest management. They established a system where this knowledge could be stored in a database that facilitates the organization and browsing of the knowledge over the World Wide Web (Thomson, 2000).

21. Justification with reference to current status *(in terms of technical/economic/societal aspects)*:

The ability to access multiple resources and rely on different modalities of land use contributes to their capacities to manage for local-level climate change and disaster risks. They offer effective solutions to addressing climate change issues through their repository of indigenous knowledge and traditional practices since these are non-polluting and sustainable. Harnessing this knowledge is especially beneficial in the Indian context since remote areas house a considerable proportion of global tribal/rural population which presently suffers from a depleting resource base and growing levels of disaster risks. Indigenous peoples' observations contribute significantly to advancing climate science, by ensuring that assessments of climate change impacts and policies for climate change

adaptation are meaningful and applicable at the local level. Traditional systems of governance and social networks improve the ability to collectively manage diversity and share resources, while dissipating shocks and reinforcing innovative capacities.

There have been fewer efforts to integrate indigenous knowledge in the disaster response process. Hence, this project indeed develops a unique state of art approach integrating AI tools and methods and the indigenous knowledge systems of the local communities and aims to build a model that can help in showing a direction for mitigating climate-induced disaster risks and aid in developing a strategic response mechanisms to support and prepare farming communities.

Although several projects have been implemented or are being implemented both nationally and internationally on data-driven disaster management, rather than filling all the gaps, they open up a plethora of opportunities that can be leveraged for improving disaster management systems. The DISANET and NDMA projects create opportunities in ensuring seamless sharing of data and information across locations and organizations using alternate wireless channels. These networks may be used to gather a huge volume of information in the form of image, video, audio and text that may be valuable in providing situational information. However, the current systems do not have enough capabilities to collect and leverage the multimodal data generated from these heterogeneous sources for obtaining real time situational information. Further the rich data collected over a period of time would also provide valuable information that may be subsequently used for better decision support, a feature which the current systems lack. Further, the existing systems are more commercial and do not provide much scope for the researchers to exploit them. This project is targeted towards mitigating these drawbacks. It is aimed towards making a scalable and auto-upgradable platform that can be leveraged by both the administrators as well as the developers to build more robust and scalable systems.

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22. Deliverables *(To be precise and quantified in bullet form)*

1. A data collection framework for collection of disaster related data from
    1. Sensors
    2. Social Communities and cultural perceptions
    3. Digitized incident and damage reports
    4. Websites
    5. Images and videos from news and weather repositories
    6. Social media feeds like twitter and reddit
2. Technical reports on novel mechanisms for
    1. Multi-modal data fusion
    2. Creating multimodal benchmark data from heterogeneous sources
    3. Task specific information filtering from multilingual social media feeds
    4. Event identification and timeline generation from multimodal and multilingual web and social media data.
3. Benchmark datasets of
    1. Images and videos classified as informative/non-informative/death/infrastructure damage/property type
    2. Text data classified as informative/non-informative/policy/infrastructure damage/need/help/location etc.
4. An unified evaluation system for testing AI based models related to
    1. Task specific information filtering
    2. Event detection
    3. Timeline generation
5. Proof of concept deployment for a specific use case scenario
    1. Technical report on planning, development and deployment for appropriate use case
    2. Evaluation and field report
6. Trained PhD students and publications in high quality journals and conferences

23. Methodology to be adopted *(Please include flow sheet if applicable)*

We organize the project activities and goals into 7 major work packages (WP0 to WP6) that can be worked on concurrently so as to ensure its time-bound completion besides maintaining high quality of the deliverables. The objectives of each of the work packages along with the tasks and deliverables are outlined as follows:

WP0: Project Management
Objectives
This work package has the following objectives: Obj 0.1 Define, set-up and implement the management process for the project, and thereafter to coordinate and manage the project accordingly. Obj 0.2 Formulate and implement a quality assurance plan for the project deliverables as well as supported tools Obj 0.3 Carry out risk assessment and management. Define contingency plans if needed. Obj 0.4 To set up and maintain the facilities for intra-project communication.
Deliverables
D0.1: Setup of Internal Communication Facilities D0.2: Periodic activity report D0.3: Final report
WP1: Development of data collection and Ingestion Framework
Objectives
Obj 1.1 Developing IoT edge nodes for data collection Obj 1.2 Developing framework for sensor data collection Obj 1.3 Developing a framework to collect traditional knowledge from local communities Obj 1.4 Develop a setup for collecting spatial data from satellite maps and drones Obj 1.5 Developing a framework for collecting multimodal data from web sources
Deliverables
D1.1 A big data setup for collection of disaster related data from sensors, digitized incident and damage reports, websites, images and videos from news and weather repositories, social media feeds. D1.2 Design, fabrication, testing and pilot field deployment of device for multihazard early warning of landslides and floods. D1.3 Collection of ethnographic and socio spatial data from the local communities

WP2: Data Analysis through Artificial Intelligence tools and methods
Objectives
Obj 2.1 Data refinement, preprocessing and labeling of data collected from sensors, edge computing devices, traditional knowledge data, spatial data and web source data Obj 2.2 Automated techniques for multimodal data alignment and fusion for data set creation
Deliverables
D2.1 A data lake of sensor data, traditional knowledge data, spatial data, web data and social media data D2.2 Novel techniques for multimodal data alignment and fusion
WP3: Development of AI models
Objectives
Obj 3.1 Models for hazard mapping Obj 3.2 Measures of preparedness indexing Obj 3.3 AI based models for knowledge extraction, situational awareness, event identification, timeline generation Obj 3.4 Models for real time early warning
Deliverables
D3.1 Models for hazard mapping D3.2 Measures of preparedness indexing D3.3 Models for knowledge extraction from community data D3.4 Models for situational awareness and damage assessment D3.5 Models for major event identification from historical data D3.6 Techniques for disaster specific activity timeline generation

D3.7 Models for real time event detection D3.8 Models for real time early warning
WP4: Disaster Management and services
Objectives
Obj 4.1 Development of visualization and reporting tools Obj 4.2 Development of tools and framework for warning dissemination Obj 4.3 Developing APIs for data loading and object creation for model development
Deliverables
D4.1 A WebGIS portal D4.2 Visualization framework for hazards, damages, events, timelines and analytic reports D4.3 A warning dissemination system D4.4 APIs for data loading and object creation for model development
WP5: Benchmarking and Evaluation
Objectives
Obj 5.1 Identifying critical features for assessing community specific vulnerabilities and risks Obj 5.2 Identifying critical features for assessing post disaster response and recovery Obj 5.3 Identifying parameters for hazard prediction Obj 5.4 Automated model evaluation platform
Deliverables
D5.1 Reports on assessment of community specific vulnerabilities and risks D5.2 Reports on techniques and features for assessment of post disaster response and

recovery D5.3 Reports on study of the critical parameters for hazard prediction D5.4 A platform for automated evaluation of AI based models on damage and need assessment, event detection and timeline generation
WP6: Use case development and proof of concept deployment
Objectives
Obj 6.1 Development of use case scenarios and deployment of proof of concepts Obj 6.2 IPR and commercialization plans
Deliverables
D6.1 Technical report on use case development: planning and evaluation markers D6.2 Technical reports on use case findings related to hazard mapping, preparedness indexing, community based studies, damage and need assessment, and assessment of post disaster response and coordination D6.3 Report on plans for IPRs and commercialization